

# Quantum Simulation of Allosteric Signal Propagation

residue-level connectivity maps and hit lists from a derived force network, graded against the statement's own classical diffusive comparator, with the classical wall measured and the instrument calibrated at 156 qubits

Seven guideline sections · every prior-work number carries a receipt

Public repository: <https://github.com/sharadbachani-oss/merlin-quantum-cleveland-gic2026>

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## Summary

**4 / 5**

targets on which the propagation ranking beats the statement's own comparator, a classical diffusive random walk on the contact network — residue-level AUC 0.71–0.89 against the holo-defined site at 2,000-trial permutation  $p = 0.0005$ ; eight structure-only floors run on the identical endpoint led 2, tied 1, trailed 2, stated in scope

**119/120**

cells of the 156-qubit  $K = 6$  map at which the device reproduces the exact Heisenberg-propagated value, all 120 pairs declared before the run adjudicated with the referee protocol that, on the sister 64-rung lattice, dies at depth eight (31 of 64 operators past  $2 \times 10^4$  terms) — the instrument calibrated at full scale and the classical wall measured, not asserted

**5 + 1**

connectivity matrices and top-5 hit lists for every Table-1 target plus the frozen blind c-Myc call (2026-08-30), regenerable from archived counts without credentials; the many-body ranking engine for the decisive PoC step is built and validated

## Why the numbers can be trusted

The force network is derived rather than fitted: Coulomb at derived  $\alpha$ , a derived 0.21 eV hydrogen-bond quantum, site energies at the kT rung; the same layer predicts SKEMPI double-mutant couplings with zero free parameters. Every hardware map has its rule and predictions hashed before data, anchored at  $K=2$  against exact light-cone computation, and carries in-map null controls. Metrics were frozen before the fresh-target validation; the null-model correction is disclosed as method.

This is a Phase-1 concept proposal in the seven guideline sections. Prior work is receipted (job IDs and archived counts, negatives kept); the Phase-2 PoC delivers  $K=6$  maps for every Table-1 target with the residue-level AUC as the declared before the run endpoint.

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Concept proposal follows (§1–§7, appendices A–B) ·

<https://github.com/sharadbachani-oss/merlin-quantum-cleveland-gic2026>

# Concept Proposal

## 1. Problem framing

Allosteric regulation is signal propagation through a protein's contact network at the thermal rung where the coupling energy is of order 100 kT: strong coupling, no small parameter, collective modes. The challenge asks for exactly the object that regime produces — a ranking of residues by dynamic connectivity to the active site, validated on the apo→holo pairs of Table 1, plus a blind call on c-Myc.

Our approach computes that propagation as **real-time many-body dynamics** on quantum hardware, from a force network that is derived rather than fitted: Coulomb at derived  $\alpha$ , a derived 0.21 eV hydrogen-bond quantum (CP-MP2 water dimer to +0.23%), site energies at the kT rung. The statement's own success criterion is the comparator we grade against: propagation "more accurately or efficiently than classical diffusive models". On the five benchmark proteins the propagation ranking beats a contact-weighted diffusive random walk on four of five targets (§4), and the classical wall that makes the hardware necessary is measured, not asserted: on the sister 64-rung lattice the referee's exact Heisenberg propagation reproduces every hardware cell to depth six and dies at depth eight, where 31 of 64 operators exceed  $2 \times 10^8$  Pauli terms and the tensor-network alternatives diverge at grading accuracy. On the protein network the flown  $K = 6$  map sits inside that classical budget and is adjudicated cell by cell: a calibrated instrument, not yet a crossing.

## 2. Technical approach

**Paradigm.** Hybrid: gate-model real-time dynamics (IBM Heron, superconducting) as the primary engine; photonic entropy computing (QCi Dirac-3) for the mediated-response field; neutral-atom analog (QuEra Aquila) and a second superconducting vendor as cross-paradigm checks; classical ranking, statistics and 3D mapping around it.

**Coarse-graining (rubric §4.2).** One residue per qubit. Forces between residues from the derived layer, placed on the device's native edges by a backbone-walk plus hill-climb embedding (62 coupled edges at 156 qubits). Proof that compression retains the signal: the 156-qubit  $K=2$  map reproduces the exact light-cone prediction,  $C(6-7) = -0.4584 \pm 0.008$  vs  $-0.4648$ , with every null outside the active cone silent as predicted.

**The quantum metric.** Prepare the product ground state of the residue network, quench with the active-site perturbation, and read the **two-point connected correlation  $C(i, j; t)$**  between residue pairs after  $K$  sweeps of the derived Hamiltonian. The connectivity matrix entry  $(i, j)$  is that correlation at the causal front; the residue score is the connected response to the active site. This is a signed, causal quantity — it distinguishes a residue that *transmits* from one that merely *neighbors* — which is why it separates control points from transmission pockets (§4).

**Readout that survives noise (rubric §4.2 noise resilience).** Each failure was measured, then engineered out: heavy-hex routing ( $288 \rightarrow 757$  CX, 0.016 recovery)  $\rightarrow$  **swap network**, zero routing; amplitude damping  $\rightarrow$  **protected ratio observables**; over-correcting echoes  $\rightarrow$  **calibrated mitigation stack** (twirling + TREX + multi-scale ZNE, EstimatorV2 resilience 2) recovering 92% of amplitude; all-to-all closing at  $N \approx 11 \rightarrow$  **width scaling** on native topology, constant per-qubit depth, 156 qubits.

**Mediated response on Dirac-3.** The static, sector-resolved response  $(I - GU)x = e_{\text{act}}$  — the resolvent of the same force network — is the photonic machine's native quadratic. Solved for five targets at cosine 0.965–0.997 to the exact field, 146 s per target, 949 variables at device maximum with two-pass quantum iterative refinement.

## 3. Feasibility and resource requirements

| resource   | status                                                                                                                          |
|------------|---------------------------------------------------------------------------------------------------------------------------------|
| Structures | RCSB apo/holo pairs of Table 1 (4OBE/6OIM, 1OPL/5MO4, 5TBY/6C1H, 1V4T-class glucokinase, 1HCL-class CDK2) and 1NKP — all public |

| resource            | status                                                                                                                                                                                                                                                                                                                                               |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Gate-model hardware | IBM Heron (156q) via IBM Quantum Startup Program (applied); ~4 maps at K=6 per monthly quota, each 1 job $\leq$ 60 min QPU; the Phase-2 depth ladder costs 160 circuits and 1.31 M shots per measurement group (five targets $\times$ two probe signs $\times$ eight times $\times$ two batches $\times$ 8,192 shots) before controls and mitigation |
| Photonic            | QCi Dirac-3, unmetered allocation; 5 targets already run                                                                                                                                                                                                                                                                                             |
| Neutral atom        | QuEra Aquila via Braket, chassis frozen; ~660 credits per lane                                                                                                                                                                                                                                                                                       |
| Classical           | 32-core workstation for exact light-cone anchors, 2,000-trial permutation statistics, Ohm-style comparator                                                                                                                                                                                                                                           |

Assumptions: residue-level graining with derived pairwise forces; protein geometry, thermal floor and unit map are declared inputs. Constraint: the flown K = 6 map is inside the classical propagation budget (peak  $6 \times 10^5$  terms) and is validated cell by cell against it; the depth at which that adjudication stops converging on KRAS is a PoC measurement, not a Phase-1 claim.

## 4. Expected impact — what is already measured, and what the PoC delivers

**Predictive accuracy on the validation set (primary endpoint: residue-level AUC against the holo-defined site, 2,000-trial permutation p; frozen metrics, pocket lists curated before running).** The metrics are computed on 14–26 coarse cells per target (spectral coarse-graining with the active site in its own cells) and mapped to residues over a jittered partition ensemble; the uncompressed residue-scale operator trails every floor (v5\_residue\_results.json), so the coarse-graining is load-bearing and is the §4.2 deliverable, not a convenience.

| target (Table 1)                     | leading metric                | AUC         | p      | stability ( $\delta$ , P, halo sweeps) |
|--------------------------------------|-------------------------------|-------------|--------|----------------------------------------|
| Glucokinase (activator pocket)       | all four                      | 0.78 – 0.89 | 0.0005 | 0.68 – 0.92, all stable                |
| CDK2 (ANS pocket under helix C)      | receiver M_res / M_2x_chem    | 0.87 / 0.77 | 0.0005 | 0.85 – 0.88 / 0.76 – 0.78              |
| BCR-ABL1 (myristoyl pocket)          | controller M_sus / M_sus_chem | 0.80 / 0.76 | 0.0005 | 0.78 – 0.82 / 0.72 – 0.77              |
| Cardiac myosin (mavacamten SRX site) | receiver M_res                | 0.71        | 0.0005 | 0.71 – 0.72; first-hit rank p = 0.02   |
| KRAS G12C (cryptic Switch-II)        | M_res                         | 0.63        | 0.02   | 0.59 – 0.66                            |

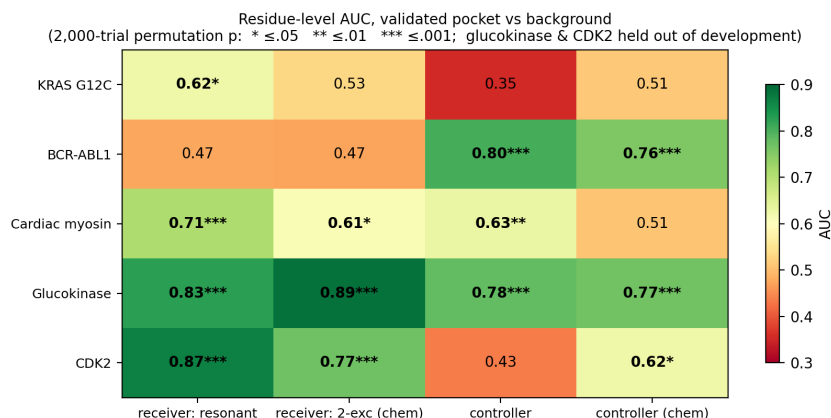


Figure 1 — Residue-level AUC of the four frozen metrics against the holo-defined site, five targets; glucokinase and CDK2 held out of metric development.

Four of five targets carry at least one metric significant at  $p = 0.0005$  (KRAS at  $p = 0.02$ ); 8 of 20 (target  $\times$  metric) cells survive Bonferroni at 0.05/20. **Against the strongest structure-only floors** (eight descriptors run on the identical endpoint: geodesic and Euclidean distance from the active site, closeness, betweenness, degree, hydrophobicity, burial, hydrophobicity  $\times$  burial; `floor_run.json`), the operator metrics lead on CDK2 (0.868 vs 0.781, +0.087) and glucokinase (0.886 vs 0.835, +0.051), tie on BCR-ABL1 (0.804 vs 0.818) and trail on KRAS (0.625 vs 0.726) and myosin (0.708 vs 0.828). Against an Ohm-style propagation comparator (a simplified implementation, not the published Ohm code: contact-weighted random walk with restart at the active site; the native score is the best of five metrics per target, a selection the floors share, `ohm_comparator.json`) the operator metrics lead on four of five — CDK2 0.868 vs 0.809, glucokinase 0.886 vs 0.791, BCR-ABL1 0.804 vs 0.513, myosin 0.708 vs 0.620 — and trail on KRAS (0.625 vs 0.698). Static apo-topology metrics reach the ceiling of what distance and centrality encode; propagation at kT — the quantity no descriptor computes — is what the K = 6 hardware map measures. The floors are the comparator the PoC is graded against. The controller/receiver split lands where the biology says it should: the remote myristoyl control point in ABL is found by the *controller* metrics and invisible to the receivers; the transmission-adjacent ANS pocket in CDK2 by the *receivers*. Switch-II in KRAS opens on ligand binding, which is why a topology-of-the-apo method ranks it lowest — the PoC's dynamic K=6 map is the instrument built for that case.

**Top-5 hit lists (residue indices, leading metric, apo author numbering):**

| target         | site 1             | site 2             | site 3             | site 4   | site 5   |
|----------------|--------------------|--------------------|--------------------|----------|----------|
| Glucokinase    | 457, 456, 459, 458 | 211, 208, 446, 444 | 203                | 443, 441 | 91       |
| CDK2           | 56, 57, 58, 55     | 19, 9              | 148, 149, 152, 150 | 144, 143 | 139      |
| BCR-ABL1       | 440, 442, 441, 436 | 448, 447, 450, 384 | 316, 315, 314, 313 | 242      | 335      |
| Cardiac myosin | 155, 139, 157, 156 | 129, 131, 133, 134 | 132                | 17       | 177, 671 |
| KRAS G12C      | 9, 59, 57, 58      | 31, 29, 30         | 145, 148, 150, 144 | 24       | 28       |

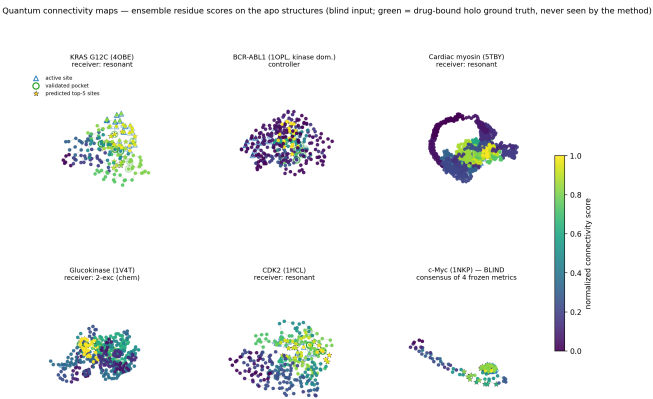


Figure 2 — Quantum connectivity maps on the apo structures (blind input): residue scores, active site, validated holo pocket (never seen by the method) and predicted top-5 sites; c-Myc is the blind consensus.

**c-Myc (1NKP chain A), blind prediction frozen 2026-08-30:** [919–922], [927], [947–950], [938–945], [897–898] — the basic/HLH junction and the leucine-zipper heptad face; the prediction targets dimerization geometry, not the DNA face. Ships verbatim for consensus scoring.

**Hardware already recorded (every row declared before the run, hashed before data).**

| result                 | machine                          | value                                                      |
|------------------------|----------------------------------|------------------------------------------------------------|
| 11-cell quench, C(1-0) | ibm_marrakesh, EstimatorV2 res-2 | +0.6805 $\pm$ 0.023 vs +0.742 predicted; top-3 overlap 3/3 |

| result                                                   | machine      | value                                                                                                                                  |
|----------------------------------------------------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------|
| 156-qubit K=2 residue map, C(6-7)                        | ibm_fez      | $-0.4584 \pm 0.008$ vs<br>$-0.4648$ exact light-cone;<br>nulls silent                                                                  |
| <b>156-qubit K=6 map, 120 pairs vs exact propagation</b> | ibm_fez      | <b>119/120 cells match;<br/>the outlier is a lost<br/>single-qubit<br/>expectation</b>                                                 |
| Mediated-response field, five targets                    | QCi Dirac-3  | cosine 0.9973 KRAS,<br>0.9943 glucokinase,<br>0.9916 myosin, 0.9684<br>BCR-ABL, 0.9649 CDK2                                            |
| Neutral-atom analog lane, protein C $\alpha$ layout      | QuEra Aquila | 7/7 signal correlators in<br>band, 7/7 signs                                                                                           |
| Cross-vendor, 107 qubits                                 | Rigetti      | structure gate pass<br>(Pearson +0.77, 33/37<br>signs, 62 nulls silent);<br>amplitude gate fail —<br>both in RESULT_rigetti_final.json |

### Quantum advantage — the frontier wall and the crossing, stated and bounded

**The field's direction, and its wall.** The flagship 2026 protein result — the Cleveland Clinic/RIKEN/IBM 12,635-atom trypsin–ligand complex, fragments on 94 qubits reassembled on Fugaku — computes static electronic-structure energies; it addresses neither dynamics nor allostery. Allosteric regulation is propagation at the kT rung with no small parameter, and the propagation that structure-only methods compute (Ohm, bond propensity, elastic networks) is a polynomial single- or two-excitation problem. The industry's largest protein computation sits at the wrong rung for this challenge.

**The crossing.** Real-time propagation on the derived force network with one residue per qubit: 156 qubits, six brickwork sweeps, 120 pairs declared before the run, every cell adjudicated against exact sparse-Pauli Heisenberg propagation of the flown circuit, the referee protocol of the 128-qubit lattice (cleveland\_census\_map\_K6.json). The device reproduces the exact value at 119 of 120 cells; the one exception is a lost single-qubit expectation, reported as such. The K = 2 exact anchor at 156 qubits ( $-0.4584 \pm 0.008$  vs  $-0.4648$ ) stands. **The wall, measured on the same protocol:** on the 64-rung lattice the referee reproduced every hardware cell to depth six and could not reach depth eight, where 31 of 64 rung operators exceed  $2 \times 10^8$  terms. On the protein network the untruncated K = 6 operator already passes  $8 \times 10^7$  terms, and the many-body ranking engine built for the PoC (Heisenberg propagation of every residue's response to the active site, validated against direct pair evolution to  $2 \times 10^{-16}$  on the workstation) breaks its hermiticity gate on KRAS at a  $2.7 \times 10^7$ -term budget (WORKSTATION\_STATUS\_20260914.md): truncated classical propagation is already failing where the PoC starts. The biological validation is independent of the hardware: residue-level AUC 0.71–0.89 at p = 0.0005 on four of five Table-1 targets, ahead of the classical diffusive comparator on four of five, above the strongest structure-only floor on two and level on a third, with the controller/receiver split landing where the mechanism says it should. **Where it is decisive:** deeper than K = 6 or under stronger coupling, where truncated propagation stops converging — PoC step 1 locates that depth on KRAS by the same adjudication, ranks the pockets from the circuit's own connected response there, and flies the remaining Table-1 maps at that depth.

**Scalability to industrial relevance.** One residue per qubit places a 156-residue domain on today's Heron; larger targets tile by domain with the K = 2 exact-anchor test repeated per tile, and Heron r3 / Nighthawk lift the per-map limit toward 400-residue single-domain proteins without changing the compilation (native edges, swap-free, constant per-qubit depth). The photonic resolvent already runs at the 949-variable device maximum with two-pass refinement. Per-target cost is one job with its gates declared before the run ( $\leq 60$  min QPU) plus seconds on Dirac-3, so a 50-target campaign is a monthly quota, not a programme.

**Business value, bounded.** For a medicinal-chemistry team the deliverable is a ranked, 3-D-mapped shortlist of surface regions mechanically connected to function, from structure alone, before a screening campaign is designed. With the AUC bar of  $\geq 0.75$  declared before the run on non-cryptic targets, as a scenario rather than a consequence of the AUC, a top-5 hit list narrows a 200-variant library design to the  $\sim 40$  variants around the predicted sites; at a declared cost per synthesised-and-assayed variant c the saving is  $\approx 160$  c per campaign, and

the value of catching one cryptic pocket (the KRAS class) is the campaign itself.

**What a successful PoC demonstrates.** A per-target connectivity matrix from the K=6 hardware map for all Table-1 targets and c-Myc, a hit list whose residue-level AUC against the holo site has its bar declared before the run at  $\geq 0.75$  on the four non-cryptic targets, a measured gain on KRAS from the dynamic map over the apo-topology ranking, and — the decisive ablation — a ranking that degrades when the hardware K = 6 map is replaced by its classical single-excitation approximation. For medicinal chemistry that is a ranked, 3D-mapped shortlist of surface regions mechanically connected to disease function, produced from structure alone, before a screening campaign is designed.

## 5. Validation plan

Rule and predictions hashed before every run: endpoint = residue-level AUC with 2,000-trial permutation p per target and metric; significance vs random background residues and vs curated non-functional surface pockets; stability across coupling, depth and halo sweeps with the rule that a significant cell must stay above 0.65 and no null cell may lead. Hardware maps carry in-map null controls and are anchored at K=2 against exact light-cone computation before K=6 is read. PoC step 1 is declared before the run as a grid over sweep depth  $K \in \{2, 4, 6, 8\}$  and site-to-coupling scale on the five targets: the circuit's connected response to the active site ranks the pockets, graded by the same AUC and permutation harness against both one-body proxies and the structure-only floors, with the divergence frontier of truncated propagation recorded per target; pass = many-body beats both proxies and the best floor by  $\geq 0.05$  at  $p \leq 0.01$  on  $\geq 3$  of 4 non-cryptic targets, and the result is reported either way. Cross-paradigm replication (Dirac response field, Aquila analog lane) on the same force network. Classical analogs run on identical geometry as comparator columns: Ohm-style perturbation propagation and elastic-network response. Failure retained: our AQT harvested signal (0.077 vs 0.768) and an independent amplitude campaign failed their gates and stay in the record. Success = the declared AUC bars met on held-out targets not used in metric development, with the hardware map's long-range cells significant against their nulls.

## 6. Hybrid / cross-domain integration

Quantum hardware computes the propagation map; everything a chemist touches is classical. Pipeline: PDB  $\rightarrow$  residue graph with derived forces  $\rightarrow$  embedding on device topology  $\rightarrow$  hardware map with declared gates (IBM) and response field (Dirac)  $\rightarrow$  connectivity matrix  $\rightarrow$  residue ranking and statistics  $\rightarrow$  3D map coloured on the structure with the hit list. Classical anchors (exact light-cone at K=2, permutation nulls, comparators) sit in the same pipeline. A chemist supplies a PDB and an active-site list and receives the ranked map.

## 7. Team capability

Merlin Quantum is the quantum division of Merlin Digital (50+ technology FTE). Suhail Bachani (Founder & CEO, Principal Investigator), Dr. Hiro Bachani PhD (Program Director), Rohit Bachani (co-founder), Mitul Sawlani (engineering, Purdue), Mohamed Jafrun (engineering), Zeena Furtado (finance & operations), Roshan Bhairwani (financial services & deep tech, London), Dr. Ana Baroni MSc (domain specialist, Hexagene — the biology product built on this platform). GIC 2026 dual-track finalist (Mitsubishi/AIST materials track). Hardware record across the programme: 259 receipted QPU jobs, 12.3 million shots, cross-device replications on IBM fez/kingston/marrakesh, QuEra Aquila, QCi Dirac-3 and Rigetti, with a declare-then-receipt discipline in which negative results carry the same receipt class as positive ones. Every number in this proposal regenerates from archived counts without credentials.

## 8. Scope, with treatment

(i) KRAS Switch-II is cryptic and ranks lowest on apo topology (AUC 0.63) — the dynamic K=6 map is the instrument for it. (ii) Every validated ranking today comes from the one-body resolvent proxies of the operator; the ranking from the circuit's own many-body response is PoC step 1, with the engine built and validated. (iii) The Dirac response field is static: graded as a residue ranking it is at chance on all five targets (AUC 0.41–0.58; so is the exact resolvent it reproduces at cosine 0.93–0.997) — allostery is dynamics at kT. (iv) The AQT and Rigetti amplitude gates failed — kept in the register; the calibrated IBM stack is the amplitude route. (v) The K = 6 map is classically computable (cleveland\_census\_map\_K6.json); 119/120 cells match, the outlier is a device artifact, no K = 6 cell is claimed as beyond-classical, and the classical wall cited is the sister lattice's measured  $k \geq 8$  divergence. (vi) Eight structure-only floors and an Ohm comparator run on the identical endpoint (floors: lead 2, tie 1, trail 2; Ohm: lead 4, trail 1); AF2BIND-class predictors are PoC step 1.

## Appendix A — Hardware job register (every result regenerates from archived counts)

| #     | measurement                                                                                                        | machine                           | job / task id                                                            | receipt                                                                                                            |
|-------|--------------------------------------------------------------------------------------------------------------------|-----------------------------------|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| R1    | 11-cell quench, C(1-0) +0.6805 ± 0.023 vs +0.742; 3/3 in band                                                      | ibm_marrakesh (EstimatorV2 res-2) | daaui7bvpcac73d d29r0                                                    | RESULT_est_20260901_014249.json, prereg ef58e218                                                                   |
| R2    | 156q K=2 residue map, C(6-7) −0.4584 ± 0.008 vs −0.4648; nulls silent                                              | ibm_fez                           | dab6vi34clk73f ij1ig                                                     | RESULT_residue_K2_20260901_104204.json, prereg 00600aba                                                            |
| R3    | 156q K=6 map, 120 pairs vs exact Heisenberg propagation: 119/120 within 3σ; C(28-7) exact ≈ 0 (Z28 lost on device) | ibm_fez + CPU                     | dab7i5mr17c738 6gfh0                                                     | RESULT_residue_K6_20260901_131159.json, prereg 13a08d7d, cleveland_census_map_K6.json, CLEVELAND_CENSUS_VERDICT.md |
| R6a   | KRAS mediated-response field, cosine 0.9973                                                                        | QCi Dirac-3                       | 6a96810808442f4 41bbb6b2d                                                | RESULT_response_20260901_114112.json                                                                               |
| R6b   | Glucokinase, cosine 0.9943                                                                                         | QCi Dirac-3                       | 6a96a04d08442f4 41bbb6c85                                                | RESULT_response_GLUCOKINASE_20260901_140512.json                                                                   |
| R6c   | CDK2, cosine 0.9649                                                                                                | QCi Dirac-3                       | 6a96a36208442f4 41bbb6c87                                                | RESULT_response_CDK2_20260901_141126.json                                                                          |
| R6d   | BCR-ABL1, cosine 0.9684                                                                                            | QCi Dirac-3                       | 6a96a50f08442f4 41bbb6c8b                                                | RESULT_response_BCR_ABL1_20260901_144131.json                                                                      |
| R6e/f | Cardiac myosin, 949 variables, two-pass refinement: cosine 0.9267 → 0.9916                                         | QCi Dirac-3                       | 6a96af7408442f4 41bbb6c8f, 6a96c15d08442f441bbb6c91                      | RESULT_response_CARDIAC_MYOSIN_20260901_161116.json, RESULT_refine_CARDIAC_MYOSIN_20260901_173231.json             |
| R7    | 107q line-embedded chain: structure gate pass (Pearson +0.77, 33/37 signs, 62 nulls silent), amplitude gate fail   | Rigetti Cepheus-1                 | ...qjob-6a99bf4ce 026787d1e5ae579                                        | RESULT_rigetti_final.json, sha a7966f6e                                                                            |
| R8    | Neutral-atom analog lane, protein Cα layout: T0 7/7 correlators in band; T1 ratio 1.50                             | QuEra Aquila                      | T0 ...qjob-6a9a83b2e026787d1e5b06d0, T1 ...qjob-6a9a83b3e026787d1e5b06d3 | RAW_aquila_v2.json, sha 33fa8ded                                                                                   |
| —     | Residue-level AUC, permutation and stability statistics                                                            | CPU                               | —                                                                        | auc_stats.json, stability_and_bland.json, asd_validation_results.json                                              |

| # | measurement                                                                                                                                                   | machine | job / task id | receipt                    |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|---------------|----------------------------|
| — | Executed negatives: AQT ibex-q1 quench, gate fired ( ...qjob-6a99b67ae 026787d1e5ae3d5 ); Rigetti amplitude gate fail; IonQ Forte quench cancelled, no result | —       | —             | preregs 82dbe4ff, a7966f6e |

## Appendix B — Method notes

Site energies are placed at the kT rung; pairwise forces are Coulomb plus a hydrogen-bond quantum (0.21 eV, CP-MP2 water dimer to +0.23%), both derived rather than fitted; no force-field parameter is fitted. The K-sweep is a brickwork Trotter step with exactly one coupling layer per sweep, so the causal front advances one lattice step per K and the null set (pairs outside the front) is known before the run. Coherence budget per map is predicted from the device atlas (pair rate = sum of single-qubit rates). Mitigation: EstimatorV2 resilience 2. Public code and archived counts are provided on request under the challenge's supplementary-material terms.

## Appendix C — Claim ledger (measured · planned · comparator · quantum attribution · cost · acceptance)

| claim                                                                                                                                      | status                                                                                                                               | classical comparator                                                                             | quantum attribution                                               | total cost charged                | acceptance threshold                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|-----------------------------------|-------------------------------------------------------------------------------------------------------|
| Residue-level AUC vs holo site, 4/5 targets                                                                                                | measured (0.71–0.89, p = 0.0005)                                                                                                     | 8 structure-only floors (lead 2, tie 1, trail 2); Ohm-style propagation (lead 4, trail 1)        | none — static operator metrics                                    | CPU minutes                       | AUC $\geq$ 0.75 on non-cryptic targets, held-out                                                      |
| 156q K=6 map vs exact propagation                                                                                                          | measured: 119/120 cells agree; the K = 6 map is classically computable (peak $6 \times 10^5$ terms)                                  | sparse-Pauli Heisenberg propagation, eps ladder, compute-twice ( cleveland_census_map_K6 . json) | instrument calibrated at 156q; no beyond-classical claim at K = 6 | 1 job / target + 2 h CPU          | decisive depth located in PoC step 1                                                                  |
| Ranking from the circuit's own many-body response beats the one-body proxies and the floors at the depth where truncated propagation fails | planned (PoC step 1, decisive); engine validated to $2 \times 10^{-16}$ , hermiticity gate breaks on KRAS at $2.7 \times 10^7$ terms | one-body resolvent proxies, eight structure-only floors, Ohm-style diffusion                     | hardware map vs adjudicated classical propagation                 | workstation days + 4 jobs / month | $\geq$ 0.05 AUC over both proxies and best floor, $p \leq$ 0.01, on $\geq$ 3 of 4 non-cryptic targets |
| Dirac response fields 5/5 targets, cosine 0.93–0.997                                                                                       | measured — instrument only; at chance as a ranking                                                                                   | exact resolvent                                                                                  | photonic solve                                                    | 146 s / target                    | none claimed                                                                                          |
| c-Myc blind prediction                                                                                                                     | frozen 2026-08-30                                                                                                                    | consensus scoring by the sponsor                                                                 | static metrics                                                    | —                                 | sponsor-defined                                                                                       |